

## QT 511: Introduction to Quantum Optics

Course Code: QT 511

Course Name: Introduction to Quantum Optics

L-T-P-C: 3-0-0-3

Prerequisites: Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model

Intended for: UG/PG/PhD

Mutual Exclusion: PH 550

Distribution: Elective PG/Elective UG

Semester: Odd/Even

Approval: 47th Senate

### Preamble

Students of this course learn (i) To quantise the electromagnetic field, (ii) The various experimental techniques in photonics, (iii) The various representations of states of light, (iv) Classical, semi-classical and fully quantum models of light-matter interaction, (v) Modelling decoherence through Master equation

### Course Content and syllabus

- **Quantization of the electromagnetic field**

- Number states, coherent states, squeezed states
- Hanbury-Brown and Twiss experiments – Photon bunching, Photon anti bunching
- Hong-Ou-Mandel interference

- **Theory of Optical coherence**

- Young's double slit experiment and first order coherence
- Coherence functions of arbitrary order
- Normal ordering, symmetric ordering and anti-normal ordering of operators
- Interferometry

- **Phase-space representations of states of light**

- Wigner distribution
- P-function and the notion of non-classicality with some examples of nonclassical states like squeezed states and their applications
- Husimi Q function

- **Light-matter interaction**

- Classical model of light-matter interaction
- Semi-classical model of light-matter interaction-
- Quantum light-matter interaction



- Rabi Model
- Jayne's-cummings model
- **Open quantum systems**
  - Fermi golden rule
  - Born-Markov Lindblad Master Equation

### Course Outcomes

Students of this course learn

1. To quantise the electromagnetic field
2. The various experimental techniques in photonics
3. The various representations of states of light
4. Classical, semi-classical and fully quantum models of light-matter interaction
5. Modelling decoherence through Master equation

### Course References

1. *Introductory Quantum Optics*, Christopher Gerry and Peter Knight, Cambridge University Press (2004)
2. *Quantum Optics*, D. F. Walls, Gerard J. Milburn, 2nd Edition, Springer (2008)
3. *Quantum Optics: An introduction*, Mark Fox, Oxford University Publishers (2006)
4. *Quantum Optics for Beginners*, Z. Ficek and M. R. Wahiddin, 1st edition, Jenny Stanford Publishing (2014)